Code No: 121AB **R15**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech I Year Examinations, October/November - 2020
MATHEMATICS-I
(Common to CE, EEE, ME, ECE, CSE, EIE, IT, ETM, MMT, AE, AME, MIE, PTM)

Time: 2 hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
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1.a) Show that the Eigen values of a skew-Hermitian matrix are purely imaginary or zero.

b) Show that the matrix $\begin{bmatrix} i & 0 & 0 \\ 0 & 0 & i \\ 0 & i & 0 \end{bmatrix}$ is unitary. Find the Eigen values and Eigen vectors. [8+7]

2.a) Reduce the quadratic form $2x_1x_2 + 2x_1x_3 + 2x_2x_3$ to canonical form.

b) Determine the values of a, b, c when $\begin{bmatrix} 0 & 2b & c \\ a & b & -c \\ a & -b & c \end{bmatrix}$ is orthogonal. [8+7]

3.a) State and verify Rolle's theorem for the function $f(x) = x^{2m-1}(a - x)^{2n}$ in (0, a).

b) Show that $h < e^h - 1 < he^h$ for $h \neq 0$. [8+7]

4. Prove that $\frac{\pi}{6} + \frac{1}{5\sqrt{3}} < sin^{-1}\left(\frac{3}{5}\right) < \frac{\pi}{6} + \frac{1}{8}$. [15]

5.a) Evaluate $\iint (x^2 + y^2)dx\,dy$ over the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ in the first quadrant by using the transformation $x = au$ and $y = bv$.

b) Evaluate $\iint r^3 dr d\theta$ over the area included between the circles $r = 2\sin\theta$ and $r = 4\sin\theta$. [8+7]

6. Evaluate $\iiint xyzdxdydz$ over the positive octant of the sphere $x^2 + y^2 + z^2 = a^2$. [15]

7. Solve by the method of variation of parameters $(D^2 - 2D)y = e^x sinx$. [15]

8. Solve the system of equation using Laplace transform
$2\frac{dx}{dt} + \frac{dy}{dt} - x - y = e^{-t}; \frac{dx}{dt} + \frac{dy}{dt} + 2x + y = e^t, x(0) = 2, y(0) = 1.$ [15]

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TUH USED PAPERS OCT/NOV 2